4633032

B.Tech. DEGREE EXAMINATION, APRIL/MAY 2014.

Third Semester

Computer Science and Engineering

DATA STRUCTURES

Time : Three hours   Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Give few examples for data structures?

2. What is a Priority Queue?

3. What is the length of the path in a tree?

4. When is a graph said to be weakly connected?

5. What are the various factors to be considered in deciding a sorting algorithm?

6. What is meant by internal sorting?

7. What is the basic idea of shell sort?

8. What is Sequential File Access?

9. Define B+ Tree indexing.

10. Define Garbage collection.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain how efficiency of an algorithm can be estimated.

Or

12. Explain pointer arrays with suitable examples.

UNIT II

13. Explain with an example the creation of a doubly linked list, insertion and deletion of nodes, and swapping of any two nodes.

Or

14. Explain the process of conversion from infix expression to postfix expression using stack.

UNIT III

15. Explain the traversals of binary tree with Examples.

Or

16. Explain the topological sort with an example.

UNIT IV

17. Explain details about Symbol Tables and hash tables.

Or

18. Explain the following sorting techniques with suitable examples.

(a) Shell sort

(b) Heap sort.

UNIT V

19. Describe the operation of B-tree using 2-3 trees.

Or

20. Discuss in detail about B+ tree indexing.

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